

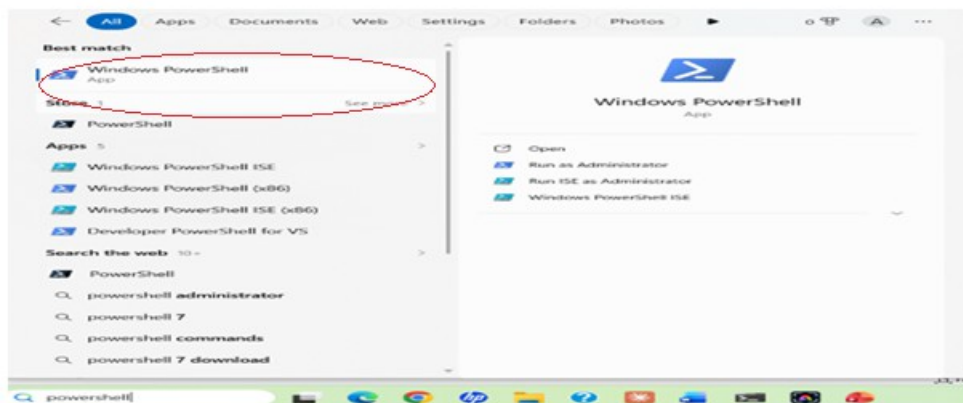
OSSP (Operating Systems And Systems Programming) - 25CS2104E

1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

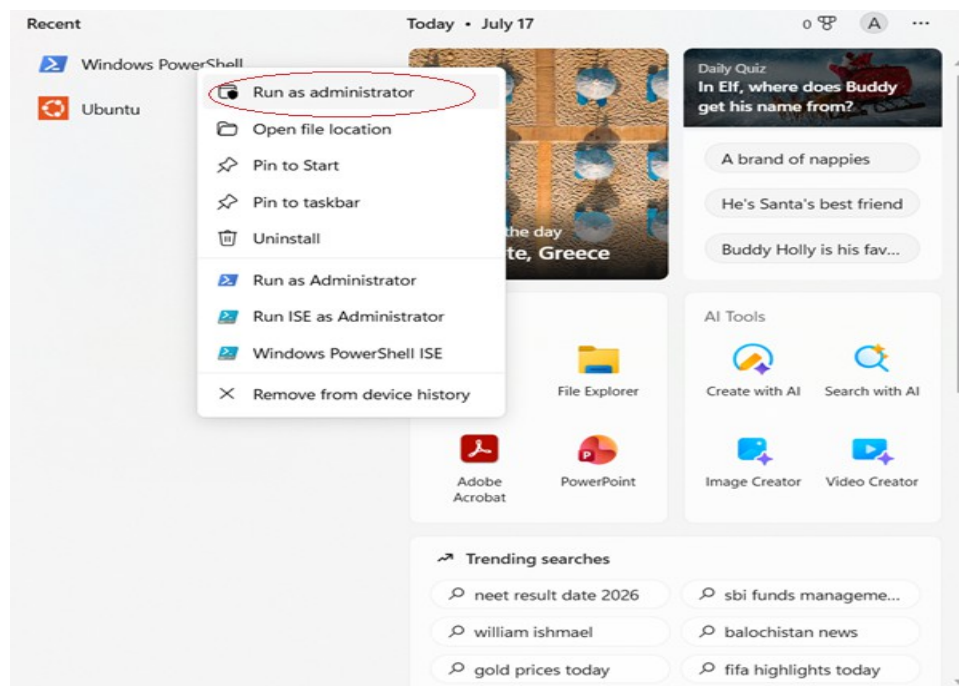
To Analyze process abstraction, execute `fork()`, understand `exec()` family, analyze parent-child relationships, inspect process tree, practice system call tracing

Step 1: Open PowerShell as Administrator

1. Press **Windows**.
2. Search for **PowerShell**.



3. Right-click **Windows PowerShell**.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\windows\system32> wsl --install
```

Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

Enter new UNIX username: neha

Step 5: Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password:****

Step 6: Update Ubuntu

```
knsat@SRITHIKA:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [422 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [495 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [97.6 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.8 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [118 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [357 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [251 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [69.0 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [83.4 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [371 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [71.7 kB]
Fetched 2985 kB in 10s (314 kB/s)
63 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Step 7: Install GCC (C Compiler)

```
knsat@SRITHIKA:~$ sudo apt install build-essential
Installing:
  build-essential

Installing dependencies:
  bzip2      g++          g++-x86-64-linux-gnu  libalgorithm-merge-perl  libfakeroot  lto-disabled-list
  dpkg-dev  g++-15      libalgorithm-diff-perl  libcrypt-dev            libfile-fcntllock-perl  make
  fakeroot  g++-15-x86-64-linux-gnu  libalgorithm-diff-xs-perl  libdpkg-perl            libstdc++-15-dev

Suggested packages:
  bzip2-doc  debian-keyring  debian-tagupload-keyring  g++-multilib  g++-15-multilib  gcc-15-doc  bzip  libstdc++-15-doc  make-doc

Summary:
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 63
  Download size: 19.3 MB
  Space needed: 73.4 MB / 1024 GB available

Continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libstdc++-15-dev amd64 15.2.0-16ubuntu1 [2920 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 g++-15-x86-64-linux-gnu amd64 15.2.0-16ubuntu1 [14.4 MB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 g++-15 amd64 15.2.0-16ubuntu1 [26.7 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 g++-x86-64-linux-gnu amd64 4:15.2.0-5ubuntu1 [968 B]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 g++ amd64 4:15.2.0-5ubuntu1 [1100 B]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 make amd64 4.4.1-3 [197 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdpkg-perl all 1.23.7ubuntu1 [289 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute/main amd64 bzip2 amd64 1.0.8-6build2 [34.3 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/main amd64 lto-disabled-list all 79 [14.2 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/main amd64 dpkg-dev all 1.23.7ubuntu1 [1062 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute/main amd64 libcrypt-dev amd64 1:4.5.1-1 [122 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 build-essential amd64 12.12ubuntu2.26.04.2 [5158 B]
Get:13 http://archive.ubuntu.com/ubuntu resolute/main amd64 libfakeroot amd64 1.37.2-1 [34.0 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute/main amd64 fakeroot amd64 1.37.2-1 [60.3 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute/main amd64 libalgorithm-diff-perl all 1.201-1 [41.8 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute/main amd64 libalgorithm-diff-xs-perl amd64 0.04-9 [11.5 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute/main amd64 libalgorithm-merge-perl all 0.08-5 [11.4 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute/main amd64 libfile-fcntllock-perl amd64 0.22-4ubuntu6 [32.8 kB]
Fetched 19.3 MB in 34s (563 kB/s)
Selecting previously unselected package libstdc++-15-dev:amd64.
(Reading database ... 48693 files and directories currently installed.)
```

Step 8: Verify the Installation

```
knsat@SRITHIKA:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Step 9: Create Your C Program

Example: nano process.c

Step 10: Compile the Program

```
knsat@SRITHIKA:~$ nano process.c
knsat@SRITHIKA:~$ gcc process.c -o process
knsat@SRITHIKA:~$ ./process
My Process ID (PID) = 4285
```